D4. Document Preprocessing, Chunking and Retrieval Index

Implementation report for retrieval over historical cybersecurity audit reports.

Field	Value
Project	Development of AI-Based Cyber Security Audit Assessment Against Benchmark
Scope	Objective 3; secure local/air-gapped RAG proof of concept
Report date	28 August 2026
Evidence rule	Implemented, tested and untested claims are labelled separately.

Data-to-evidence architecture

The retrieval layer begins with evidence preservation, not embeddings. The JSON parser flattens nested findings while retaining field paths and report identity, so a chunk can be linked back to the report structure used by an auditor. Image and figure fields are deliberately excluded because this submission does not claim visual understanding. [E6][E8]

The index combines dense vectors for semantic paraphrase with BM25 for exact security vocabulary. This is a strong fit for audit reports, where a question may use general language while the underlying report contains exact values such as CWE identifiers, severity labels, dates or remediation status. [E8][E9]

Chunking investigation

Configuration	Chunks	Accuracy	Macro-F1	Interpretation
Fixed 240 / overlap 30	24	62.50%	0.4190	More fragments; no accuracy gain
Fixed 420 / overlap 60	14	75.00%	0.5253	Best tested Qwen2.5 setting
Fixed 600 / overlap 90	11	62.50%	0.4889	Larger context; parse/review issues
Contextual 420 / overlap 60	22	37.50%	0.3294	Not selected

Implemented processing path

Input reports are parsed to provenance-labelled text. Recursive JSON objects and lists are flattened while retaining field paths; duplicate leaf rows are removed; keys containing image or figure are intentionally excluded from this text-only scope. [E8]

Chunks have stable identifiers and source context. The retrieval path combines dense FAISS search and BM25, fuses rankings, and can apply a reranker before the prompt receives final evidence. [E8]

Configuration used in the two-report run

Component	Observed configuration	Reason / boundary
Inputs	2-report.json and 4-report.json	Two source reports; intermediate Markdown was not indexed [E4][E6]
Chunking	structure_aware; max 420 tokens; overlap 60; minimum 80	22 chunks from two reports [E4]
Embedding	Qwen3-Embedding-0.6B; CUDA:0	GPU retrieval while generation uses local Qwen3-8B [E4]
Retrieval	hybrid; top_k=8; candidates_k=60; parent-child and graph enabled	No isolated ablation for parent-child/graph [E4]
Reranking/verification	reranker and local claim verifier disabled	6 GB RTX 4050 resource trade-off [E4]

Routing and evidence controls

The adaptive route identifies exact, local and aggregate question modes. Citation IDs must belong to retrieved evidence before a response is considered grounded. Prompt-injection-like report text is detected as untrusted source content. [E7][E8]

Late interaction and visual retrieval integration points exist but their model weights and visual evaluation have not been completed; neither is claimed as operational evidence. [E8]

Verification

JSON-path preservation and image-context exclusion passed a focused unit test. The offline mock test passed document parsing, chunking, hybrid indexing and metric generation without model downloads. [E7]

The two-report logged run completed its 16 questions with n_errors=0 and recorded its exact configuration. [E4]

Conclusion

The retrieval index is functional for supplied text/JSON evidence. It is not yet validated for scanned PDFs, screenshots, tables requiring visual reasoning, multilingual reports, or unseen agencies.

Evidence lineage reviewed

Stage	Concrete output	Reviewer check
Input	2-report.json and 4-report.json	Only supplied JSON reports were indexed; intermediate Markdown was excluded.
Parse	Provenance-labelled paths	Nested fields remain attributable to report structure; image/figure keys excluded.
Chunk	22 structure-aware chunks	IDs are retained in run configuration and question-level records.
Retrieve	Top-8 hybrid evidence candidates	Returned IDs/scores are available for every question.
Generate	Schema-bound response	Cited IDs must be in the retrieved set. [E4][E5]

Configuration decision

The real-report pipeline uses a structure-aware 420-token/60-token overlap configuration because audit evidence is naturally organised as nested findings and recommendations. That arrangement preserves report context while allowing question-level provenance. [E4][E8]

The separate fixed-window experiment is valuable, not contradictory: 420/60 fixed chunks achieved 75.00% answer accuracy on the Qwen2.5 control, versus 62.50% for structure-aware chunks. It is not automatically substituted into the Qwen3 run because the existing gold evidence IDs are tied to structure-aware boundaries; changing them without re-authoring labels would corrupt the retrieval comparison. [E10]

Repository evidence cited

- [E4] results/production_real_reports/1787846335_de0ded86a7db/run_config.json and run.log: two-report production-parser run configuration and execution log.
- [E6] sample_data/{2-report.json,4-report.json,real_reports_benchmark_questions.json}: source reports and eight source-derived questions per report.
- [E7] Verification executed 28 August 2026: python -m unittest test_json_parser.py test_adaptive_rag.py test_security_gates.py (6 tests passed) and python test_harness_mock.py (offline mock end-to-end passed).
- [E8] src/{docling_parser.py,chunking.py,embed_index.py,pipeline.py,evaluation_harness.py,adaptive_rag.py,prompts.py,icl.py}: implementation inspection.

- [E10] results/final_validation/VALIDATION_SUMMARY.md and its cited run directories: controlled local LLM, ICL, prompt, retrieval, CRAG, chunking and chunk-parameter comparisons completed 28 August 2026.